

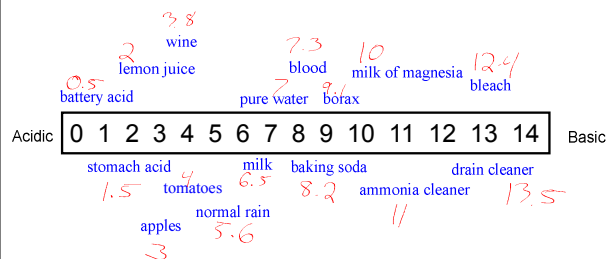
The pH Scale

The pH of a solution is a measure of how acidic or how basic it is.

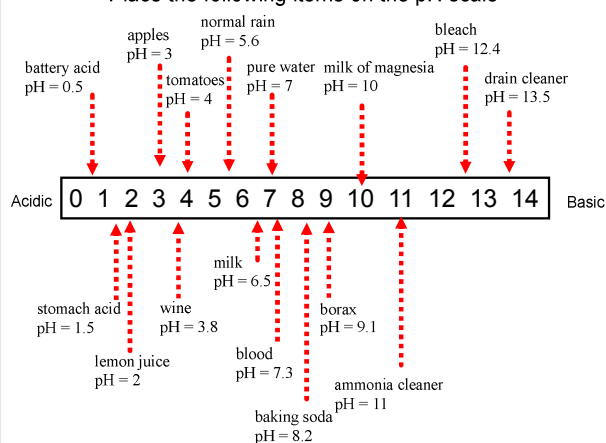
The pH scale is a scale ranging from 0-14, 0 being the most acidic and 14 being the most basic.

If a solution is neither acidic nor basic, it is said to be neutral. Neutral solutions have a pH of 7. Therefore, *pure water* has a pH of 7.

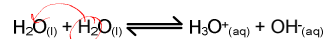
Place the following items on the pH scale



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Water is not strictly H_2O molecules; it also contains H_3O^+ and OH^- ions at low concentrations due to ionization.



Pure water has these ions in concentrations of:

$$[\text{H}_3\text{O}^+] = [\text{OH}^-] = 1.0 \times 10^{-7} \text{ M}$$

pH (power of Hydronium ions) is a measure of H_3O^+ ions with a solution, and is calculated as follows:

$$\text{pH} = -\log[\text{H}_3\text{O}^+]$$

log means "power of ten"

ex) $\log 100 = \log 10^2 = 2$

$\log 1 \times 10^5 = 5$

$\log 0.001 = -3$

$\log 1 \times 10^{-4} = -4$

$\log 25 = 1.4$

Therefore,

Ex 1) What is the pH of a solution with $[\text{H}^+] = 1.0 \times 10^{-10}$

$$\text{pH} = -\log[\text{H}^+] = -(-10) = 10$$

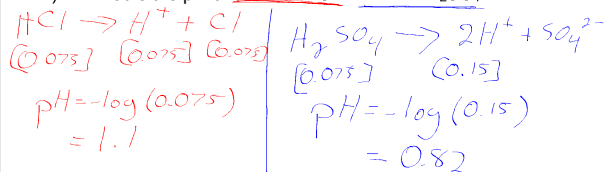
Ex 2) What is the pH of a solution with $[\text{H}^+] = 2.5 \times 10^{-4}$

$$\text{pH} = -\log[\text{H}^+] = -(-3.6) = 3.6$$

Ex 3) What is the $[H^+]$ if the pH of a solution is 5.4?

$$\begin{aligned} pH &= -\log [H^+] \\ 5.4 &= -\log [H^+] \\ [H^+] &= 10^{-5.4} \\ &= 3.98 \times 10^{-6} \\ &\approx 4.0 \times 10^{-6} \end{aligned}$$

Ex 4) What is the pH of 0.075 M HCl? 0.075 M H_2SO_4 ?



Similarly, we could discuss pOH (power of hydroxide ions)
 - a measure of OH^- ions with a solution:

$$pOH = -\log[OH^-]$$

$$pH + pOH = 14$$

Therefore,

Ex 5) What is the pOH of a solution with $[OH^-] = 1.8 \times 10^{-4}$

$$\begin{aligned} pOH &= -\log(1.8 \times 10^{-4}) \\ &= 3.7 \end{aligned}$$

Ex 6) Find the pH & pOH of a 0.0050 M $NaOH$ solution

$$\begin{aligned} [OH^-] &= 3(0.0050) \\ &= 0.0150 \\ pOH &= -\log(0.0150) \\ &= 1.8 \end{aligned}$$

$pH + pOH = 14$
 $pH = 14 - 1.8$
 $= 12.2$